

Predicting Extreme Climate Behavior with Machine Learning

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Today's goals



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Module 1

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Unsupervised Learning: Dimensionality Reduction

Module 2

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Unsupervised Learning: Clustering

K-Means Clustering

✔ Introduction to K-Means Clustering
Video • 13 min

✔ K-Means Clustering Mathematical
Video • 14 min

✔ K-Means Clustering Theoretical
Reading • 30 min

✔ K-Means Clustering Extension
Reading • 45 min

✔ K-Means Clustering
Graded Assignment • Grade: 100%

Module Assignment

✔ What is Clustering: Notebook Walkthrough
Video • 10 min

✔ What is Clustering: Notebook Walkthrough
Lab • 30 min

✔ Cluster Analysis
Reading • 30 min

✔ Clustering and Trend Analysis of Global Extreme Droughts from 1900 to 2014
Reading • 20 min

● Clustering
Programming Assignment • 1h

Module 3

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Supervised Learning: Regressions

Module 4

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Supervised Learning: Logistic Regression, Decision Trees, and SVMs

Module 5

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Supervised Learning: Neural Networks

Clustering and Trend Analysis of Global Extreme Droughts from 1900 to 2014

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The following study examines the spatial and temporal patterns of severe global droughts from 1900 to 2014, using the Palmer Drought Intensity Index (PDSI). By applying K-means clustering to extreme negative PDSI values, the research identifies significant drought events and trends over time. The findings reveal that droughts from 1950 to 1980 were less intense compared to other decades. Notably, 2012 saw the most extensive and severe droughts, particularly in Canada. The study also highlights a significant increase in the areas experiencing extreme droughts since 1980, with regions such as northern Canada, central U.S., southwest Europe, and southeast Asia being most affected.

Please read:

[Clustering and Trend Analysis of Global Extreme Droughts from 1900 to 2014](#) [↗](#) (external link; opens in a new tab)

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https://www.coursera.org/learn/predicting-extreme-climate-behavior-with-machine-learning/supplement/VnVVz/clustering-and-trend-analysis-of-global-extreme-droughts-from-1900-to-2014

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